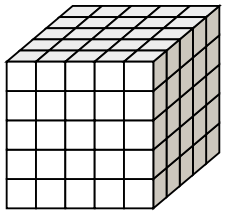
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 46754

Part A: Each question is worth 3 points

1. A cube with sides of 5 cm is painted grey all over and then cut into small cubes with sides of 1 cm.



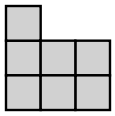
How many of the small cubes have exactly 2 painted faces?

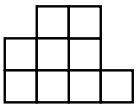
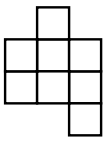
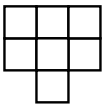
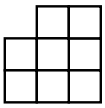
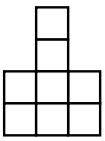
A. 60 ***B.*** 54 ***C.*** 38 ***D.*** 36 ***E.*** 18

2. Mia, who is older than Sara by 1 year and 3 days, was born on March 31, 2008.
When was Sara born?

A. April 3, 2007 ***B.*** April 3, 2009 ***C.*** April 4, 2009
D. April 3, 2010 ***E.*** March 31, 2009

3. Which of the five pieces fits together with the grey piece to make a rectangle? Pieces may be turned and flipped over.



A. 
B. 
C. 
D. 
E. 

4. The digits from 1 to 9 are written on 9 cards. Nina has 9, 5 and 4; Sara has 8, 6 and 2; Omar has 7, 3 and 1. Each of them uses some of +, –, ×, ÷ (and brackets) and each of their own cards exactly once.
Who **cannot** get 21 as the result?

A. Nina ***B.*** Sara ***C.*** Omar
D. All can get 21 ***E.*** Nina and Sara

5. Sara made a tunnel out of identical cubes (fig. 1). Then Sara took it apart and built a complete pyramid (fig. 2) with some of the cubes. Nothing is hollow inside.

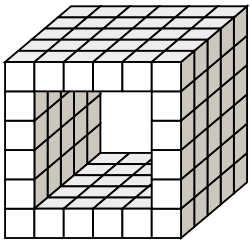


fig. 1

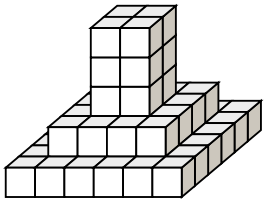
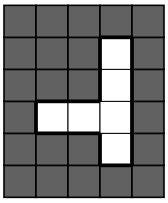


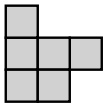
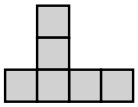
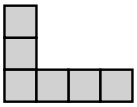
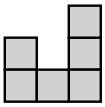
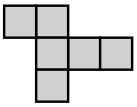
fig. 2

How many cubes from the tunnel were **not** used for the pyramid?

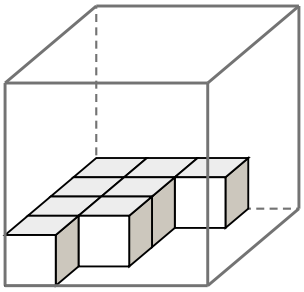
A. 56 ***B.*** 36 ***C.*** 34 ***D.*** 27 ***E.*** 16

6. Which piece covers the empty area exactly? The pieces may be turned and flipped over.



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

7. Leo puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube.



How many cubes are in the box?

A. 10 ***B.*** 9 ***C.*** 8 ***D.*** 7 ***E.*** 6

Part B: Each question is worth 4 points

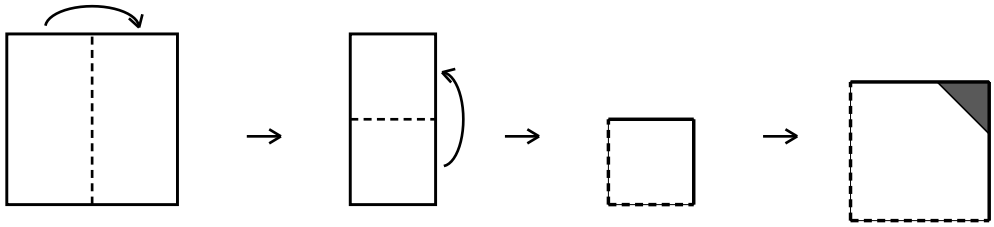
8. Leo builds a brick 6 cubes long and 4 cubes tall from red and blue cubes. The outside is completely red and the inside is completely blue. Leo used exactly 16 blue cubes. How many cubes deep is the brick?

A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 8

9. Nina thinks of a number, divides by 4, then multiplies by 3, then adds 8, and gets 14. Which number did Nina think of?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

10. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). The dark part is cut off and falls on the table. It is four layers of paper.



How many separate pieces of paper are on the table?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 8

11. The pages in a book are numbered from 1 to 126. How many times is the digit 6 written?

A. 23 ***B.*** 24 ***C.*** 26 ***D.*** 27 ***E.*** 28

Part C: Each question is worth 5 points

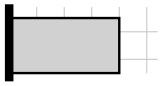
12. A red lamp flashes every 4 seconds and a blue lamp flashes every 6 seconds. They flash together right at the start. How many times do they flash **together** in the next 84 seconds? (Count a flash exactly at second 84, but not the one at the start.)

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 14 ***E.*** 35

13. Nicolas, Manon, Omar, Mia and Sofia each play a different instrument: the piano, the flute, the trumpet, the xylophone and the guitar. Sofia doesn't play the flute. Omar plays with sticks. Manon plays an instrument with strings. Mia does not blow into the instrument. Which statement is **not** true?

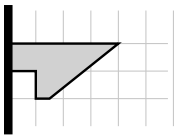
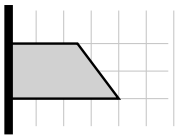
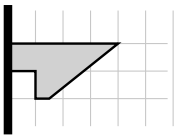
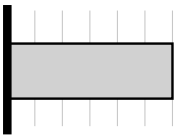
A. Mia plays the piano ***B.*** Omar plays the xylophone ***C.*** Manon plays the guitar
D. Manon plays the trumpet ***E.*** Nicolas plays the flute

14. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Leo draws the shadow at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Four of the drawings are wrong.



the flag, flat

Which drawing can be right?

A.  ***B.***  ***C.***  ***D.***  ***E.***

15. Iris paints a flag with three stripes: top, middle and bottom. Iris has 5 colours (white, yellow, red, blue and black). All three stripes must be different colours.
How many different flags can Iris paint?

A. 60 ***B.*** 20 ***C.*** 15 ***D.*** 12 ***E.*** 10

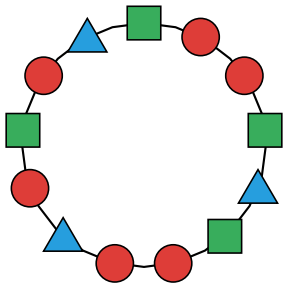
16. How many three-digit numbers have digits that add up to 7?

A. 56 ***B.*** 34 ***C.*** 29 ***D.*** 28 ***E.*** 27

17. Finn's new rug is 3 m 5 cm long. Four of these lengths are the same as that.
Which one is **not** the same?

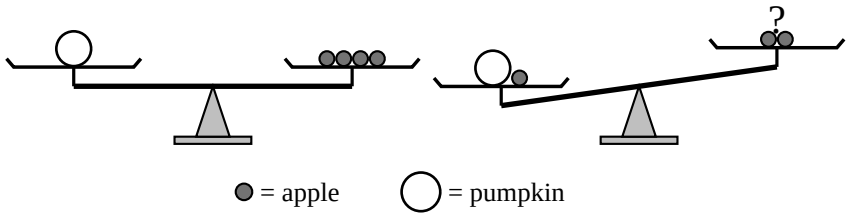
A. 3050 mm ***B.*** 305 cm ***C.*** 3 m 50 mm ***D.*** 35 cm ***E.*** 30 dm 5 cm

18. Tom has this bracelet with circles (red), squares (green) and triangles (blue) and wants to take away three **neighbouring** stones, one of each shape. In how many different places of the bracelet is this possible?



A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 10

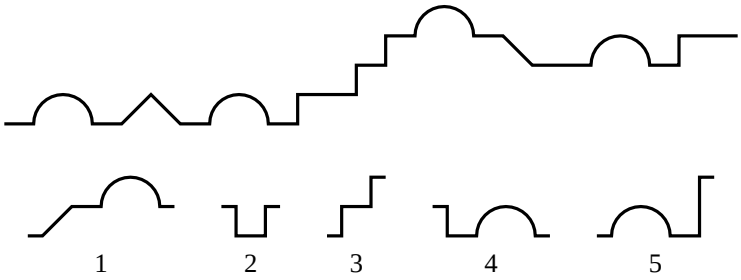
19. The first balance is level. On the second balance there is 1 pumpkin and 1 apple on the left and 2 apples on the right.



How many apples must be added to the right plate to make it level?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 5 ***E.*** 7

20. This is the outline of a town seen from the river at sunset.



How many of the pieces 1–5 below belong to the line?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

Answer Key

1.	**D**	**2.**	**B**	**3.**	**D**	**4.**	**E**	**5.**	**B**	**6.**	**B**
7.	**C**	**8.**	**B**	**9.**	**A**	**10.**	**D**	**11.**	**A**	**12.**	**B**
13.	**D**	**14.**	**B**	**15.**	**A**	**16.**	**D**	**17.**	**D**	**18.**	**A**
19.	**C**	**20.**	**A**								

Solutions

1. D (36) — They sit on the edges, but not at the corners. 12 edges with 5 – 2 = 3 such cubes each: 12 × 3 = 36. [set-03/by_level/Grade_03-04/2007 Q17 · two_faces]

2. B (April 3, 2009) — Older means born earlier, so Sara was born 1 year and 3 days **after** March 31, 2008: 1 year later is March 31, 2009, then 3 more days: April 3, 2009. [set-03/by_level/Grade_03-04/2007 Q12 · younger_birthday]

3. D — The grey piece and the right piece together fill a 3 × 5 rectangle: 15 – 7 = 8 squares, in exactly the shape of the gap. Piece D (turned) fills it; the others have a square too many, too few, or in the wrong place. [set-03/by_level/Grade_03-04/2007 Q8 · complete_rectangle]

4. E (Nina and Sara) — Omar: 1 × 3 × 7 = 21. Nina: with 9, 5 and 4 nothing gives 21 (the closest are 18 = 4 + 5 + 9 and 25 = (9 – 4) × 5); Sara: with 8, 6 and 2 nothing gives 21 (the closest are 20 = 2 × 6 + 8 and 22 = 2 × 8 + 6). [set-03/by_level/Grade_03-04/2002 Q21 · who_cannot]

5. B (36) — Tunnel: 20 cubes in the front, 5 long: 20 × 5 = 100. Pyramid: 6 × 6 + 4 × 4 + 2 × 2 × 3 = 64. Left over: 100 – 64 = 36. [set-03/by_level/Grade_03-04/2002 Q20 · tunnel_to_pyramid]

6. B — The empty area has 6 squares; all pieces do too, so look at the shape. Turn piece B and it fits exactly; each of the others has one square in a different place. [set-03/by_level/Grade_03-04/2003 Q18 · one_piece]

7. C (8) — Count each column from the bottom: 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 = 8. [set-03/by_level/Grade_03-04/2007 Q14 · count_inside]

8. B (4) — The blue inside is 4 long and 2 tall, so one layer of it has 4 × 2 = 8 cubes. 16 ÷ 8 = 2 layers of blue, plus the red front and back: 4. [set-03/by_level/Grade_03-04/2003 Q9 · how_deep]

9. A (8) — Go backwards: 14 – 8 = 6, 6 ÷ 3 = 2, 2 × 4 = 8. [set-03/by_level/Grade_03-04/2007 Q5 · think_of_a_number]

10. D (4) — The cut is at a real corner of the paper: the four layers are not joined there, so it gives 4 separate pieces (the four corners). [set-03/by_level/Grade_03-04/2007 Q24 · count_pieces]

11. A (23) — As the last digit: 13 times; as the tens digit: 10 times. Total 23. [set-03/by_level/Grade_03-04/2003 Q16 · count_one_digit]

12. B (7) — Red flashes at 4, 8, 12, ... and blue at 6, 12, 18, ... The first time both flash is at second 12 (not 4 × 6 = 24), then every 12 seconds. 84 ÷ 12 = 7. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · lamps]

13. D (Manon plays the trumpet) — The only possible matching: Nicolas – flute; Manon – guitar; Omar – xylophone; Mia – piano; Sofia – trumpet. [set-03/by_level/Grade_03-04/2002 Q18 · which_not_true]

14. B — Check two things: the pole edge must be covered all the way, and nothing may reach more than 4 squares from the pole. Only picture B passes both checks. [set-03/by_level/Grade_03-04/2002 Q15 · only_one_right]

15. A (60) — Top stripe: 5 choices, middle: 4, bottom: 3. 5 × 4 × 3 = 60. [set-03/by_level/Grade_03-04/2002 Q22 · stripes]

16. D (28) — Count by the first digit: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28. [set-03/by_level/Grade_03-04/2007 Q4 · count_with_sum]

17. D (35 cm) — 3 m 5 cm = 305 cm = 3050 mm. But 35 cm = 35 cm. [set-03/by_level/Grade_03-04/2003 Q10 · not_equal]

18. A (6) — Go once around the bracelet and look at each group of 3 neighbours. Don't forget the groups across the top: 2 of them. In all 6. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · count_places]

19. C (3) — One pumpkin = 4 apples, so the left plate is worth 1 × 4 + 1 = 5 apples. The right plate has 2, so add 5 – 2 = 3. [set-03/by_level/Grade_03-04/2002 Q9 · what_to_add]

20. A (1) — Only piece 3 fits. Not: 1 (it is a mirror image of a part of the line); 2 (it steps down where the line steps up); 4 (it steps down where the line steps up); 5 (its step is taller than in the line). [set-03/by_level/Grade_03-04/2002 Q7 · how_many_fit]